FINANCIAL FRAUD DETECTION PROJECT

Zidio Internship Project

Developed by

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Project Duration:

One month

(3-10-26 to 3-11-26)

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Project Overview:

The Financial Fraud Detection Model with Dashboard is a Machine Learning and Data Analytics project designed to identify fraudulent transactions in financial datasets. The project uses Python libraries such as Pandas, NumPy, Matplotlib, Seaborn, and Scikit-learn to perform data processing, exploratory data analysis, visualization, and model development.

The project uses a credit-card transaction dataset containing 284,807 rows and 31 columns, including transaction time, amount, anonymized features and a Class column indicating whether a transaction is fraudulent. The objective is to analyse transaction patterns and evaluate machine learning models for detecting suspicious transactions.

The project outline additionally describes interactive dashboards, ETL using SQLite, real-time transaction monitoring, and automated fraud alerts as components of the proposed system. These features can help present fraud patterns and support financial risk analysis.

Project Objectives:

The main objective of the Financial Fraud Detection Model with Dashboard is to analyse credit-card transaction data and use machine learning techniques to identify potentially fraudulent transactions.

1. Fraud Detection:

- Identify fraudulent and legitimate transactions using machine learning classification algorithms.
- Analyse transaction patterns to distinguish suspicious activities from normal transactions.
- Clean and prepare the credit-card transaction dataset for analysis,
- Address class imbalance to improve fraud detection performance.

2. Machine Learning Model Development

- Implement and evaluate Logistic Regression, K-Nearest Neighbors (KNN), Support Vector Classifier , Decision Tree, and Random Forest models.

3. Model Performance Evaluation

- Evaluate models using accuracy, precision, recall, F1-score, and ROC-AUC.

- Analyse confusion matrices and classification reports to understand fraud detection errors.

4. Data Visualization and Analysis.

- Use charts and dimensionality reduction techniques such as PCA and t-SNE to explore patterns in the dataset.

5. Dashboard and Reporting

Present fraud analysis and model evaluation results through visual dashboards.

Use Power BI and Stream lit as proposed dashboard platforms for dashboards.

3. Architecture Overview:

The architecture of the Financial Fraud Detection Model with Dashboard describes the flow of transaction data from data loading and preprocessing to machine learning model evaluation and dashboard reporting.

3.1 System Architecture:

- ❖ Data Input: Load the creditcard.csv dataset containing transaction details, anonymized features, transaction amounts, and fraud labels.
- ❖ Data Preprocessing: Inspect and prepare the dataset, scale relevant features
- ❖ Exploratory Data Analysis: Analyse transaction distributions, fraud patterns, correlations, and data characteristics using Python visualizations.
- ❖ Dashboard and Reporting: Present analytical findings and model performance through proposed Power BI or Stream lit dashboards.

Architecture Flow:

Data Sources

↓

Data Cleaning & Preprocessing

↓

Exploratory Data Analysis & Feature Engineering

↓

Machine Learning – Fraud Detection

↓

Fraud Analysis & Risk Identification

↓

Business Intelligence Layer

↓

Power BI and Streamlit Dashboard

↓

Fraud Insights and Risk Analysis Reports

4. Project Execution Timeline:

The Financial Fraud Detection Model with Dashboard is planned over four weeks, from 7 October 2026 to 3 November 2026, covering data preparation, machine learning development, fraud analysis, dashboard development, testing, and documentation.

Week 1 – Planning & Data Preparation (7 October – 13 October 2026)

- Project planning and requirement analysis
- Credit-card transaction dataset collection
- Data loading and initial inspection
- Data cleaning and quality checks

Week 2 – Feature Engineering & ML Development (14 October – 20 October 2026)

- Exploratory Data Analysis (EDA)
- Feature scaling using RobustScaler
- Handling class imbalance

Week 3 – Fraud Analysis & Model Evaluation (21 October – 27 October 2026)

- Training classification algorithms
- Comparing Logistic Regression, KNN, SVC, Decision Tree, and Random Forest
- Evaluating accuracy, precision, recall, F1-score, and ROC-AUC

- Analyzing confusion matrices and fraud patterns

Week 4 – Dashboard, Testing & Documentation (28 October – 3 November 2026)

- Developing Power BI and Streamlit dashboards
- Visualizing fraud analysis and model performance
- Testing and reviewing analytical results
- Preparing project documentation and final reports

Streamlit Dashboard and Its Key Features:

The Streamlit Dashboard for the Financial Fraud Detection Model provides an interactive interface to analyze credit-card transactions, visualize fraud patterns, and present machine learning model evaluation results. It is designed to make fraud-related insights easier to understand through charts, metrics, and data tables.

1. Dashboard Overview:

- Display total transactions.
- Show fraudulent and legitimate transaction counts.

2. Fraud Analysis:

- Visualize fraudulent versus legitimate transactions.
- Identify patterns in the transaction dataset.

3. Transaction Data Explorer:

- Display transaction records in an interactive table.
- Separate fraudulent and legitimate transactions using the Class column.

4. Interactive Filtering:

- Allow users to select transaction categories and adjust available filters.
- Update supported visualizations according to selected filters.

5. Reporting and Insights:

- Summarize fraud patterns and model evaluation findings.
- Support data-driven analysis of suspicious transactions.

Technical Highlights:

1. Data Processing

- Loaded and analyzed the credit-card transaction dataset using Pandas and NumPy.
- Performed data inspection and preprocessing to prepare the data for machine learning.

2. Machine Learning

- Applied Logistic Regression, K-Nearest Neighbors (KNN), Support Vector Classifier (SVC), Decision Tree, and Random Forest algorithms.
- Developed a classification workflow to distinguish fraudulent and legitimate transactions.

3. Model Evaluation

- Evaluated classification models using accuracy, precision, recall, F1-score, and ROC-AUC.
- Used confusion matrices and classification reports to analyze model performance.

4. Streamlit Dashboard and Visualization

- Planned an interactive dashboard to display fraud analysis and model performance.
- Used Python-based visualizations and proposed Power BI integration for reporting and data-driven decision-making.

DAX Measures and Formulas Used:

The Financial Fraud Detection Model with Dashboard can use DAX measures in Power BI to calculate transaction counts, fraud rates, transaction amounts, and other key performance indicators.

- Total Transactions = COUNTROWS('Transactions')
- FraudulentTransactions=CALCULATE(COUNTROWS('Transactions'),'Transactions'[Class] = 1)

- Legitimate Transactions =CALCULATE(COUNTROWS('Transactions') 'Transactions'[Class] = 0)

- Fraud Percentage =DIVIDE ([Fraudulent Transactions],[Total Transactions], 0) * 100

- Total Transaction Amount =SUM('Transactions'[Amount])

- Fraud Transaction Amount=CALCULATE(SUM('Transactions'[Amount]),'Transactions'[Class] = 1)

- Legitimate Transaction Amount =CALCULATE(SUM('Transactions'[Amount]), Transactions'[Class] = 0)

- Average Transaction Amount = AVERAGE('Transactions'[Amount])

- Fraud to Legitimate Ratio = DIVIDE ([Fraudulent Transactions], [Legitimate Transactions], 0)

- Fraud-Free Transaction Percentage =DIVIDE([Legitimate Transactions], [Total Transactions], 0) * 100

Technologies Used:

1. Python - Main programming language used for data analysis, preprocessing, visualization, and machine learning model development.

2. Pandas and NumPy - Used for loading, manipulating, cleaning, and processing credit-card transaction data.

3. Matplotlib and Seaborn - Used to create charts, distribution plots, and correlation heatmaps for exploratory data analysis.

4. Scikit-learn - Used for machine learning algorithms, feature scaling, model training, cross-validation, and performance evaluation.

5. Imbalanced-learn - Used to support analysis and evaluation of imbalanced classification data.

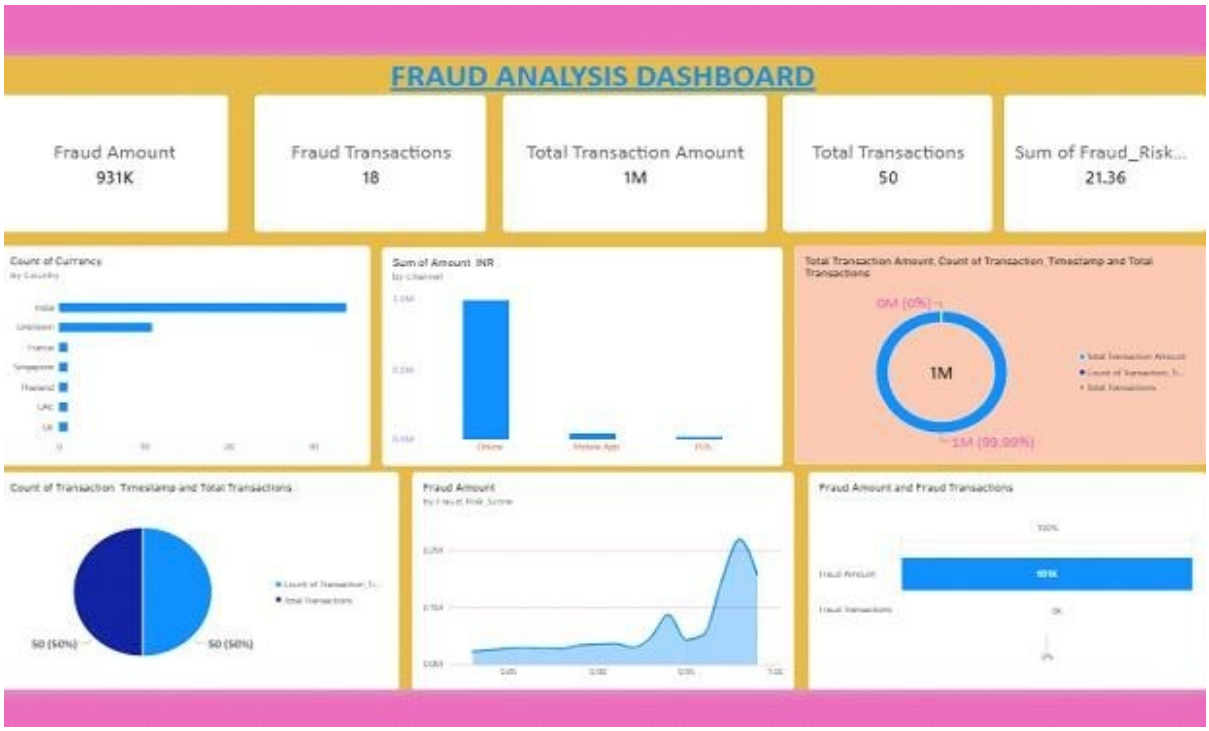
6. Jupyter Notebook - Used as the development environment for executing Python code and documenting the fraud detection workflow.

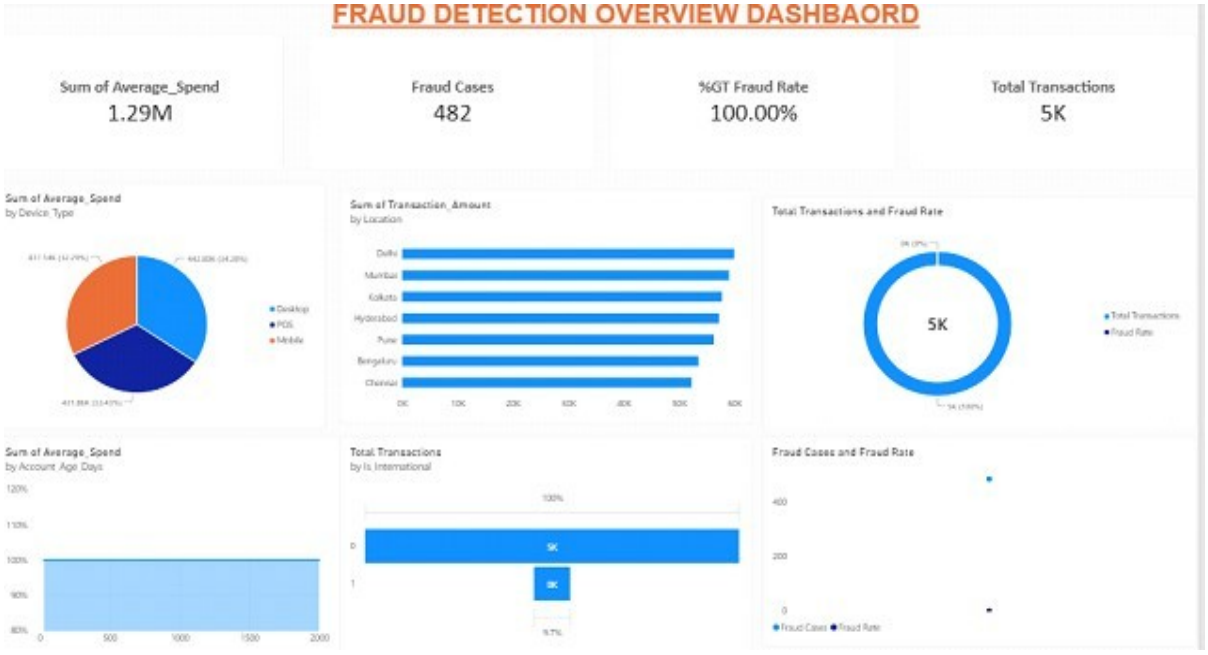
7. Streamlit - Proposed for building an interactive web dashboard to display fraud analysis and model performance.

8. Power BI - Proposed for business intelligence reporting and visualization of fraud-related KPIs using DAX measures.

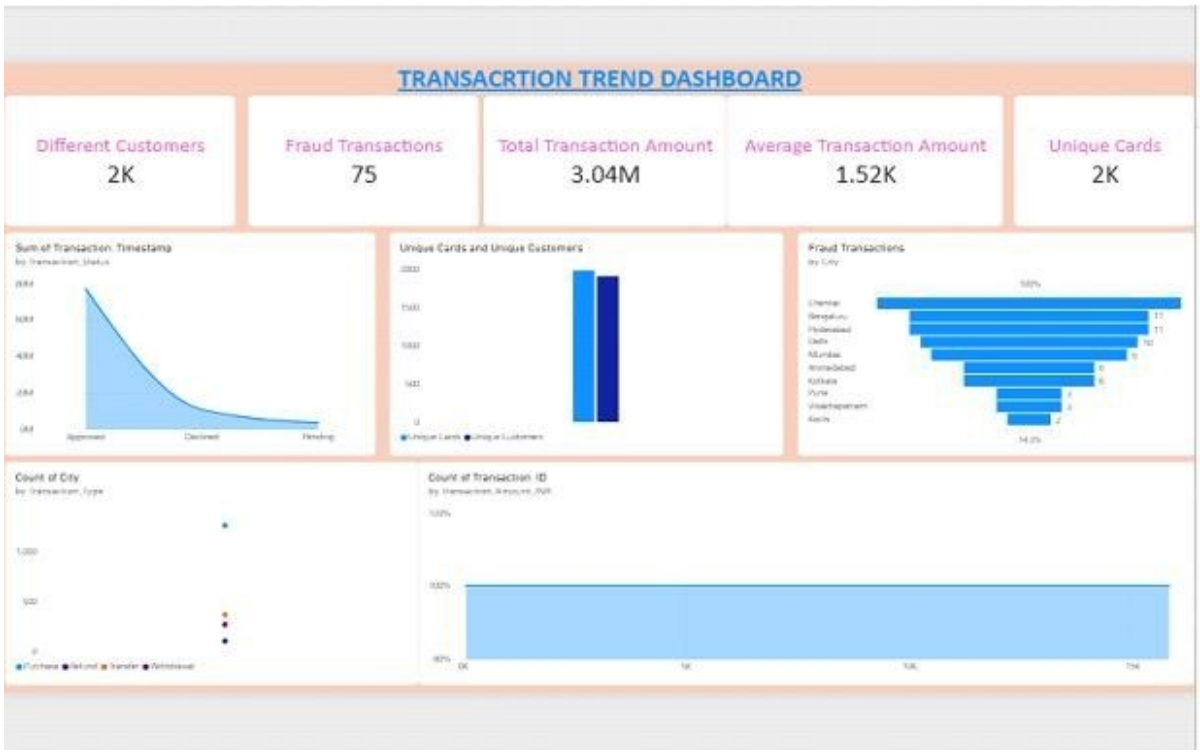
9. SQLite - Included in the project outline for storing processed transaction data through an ETL workflow.

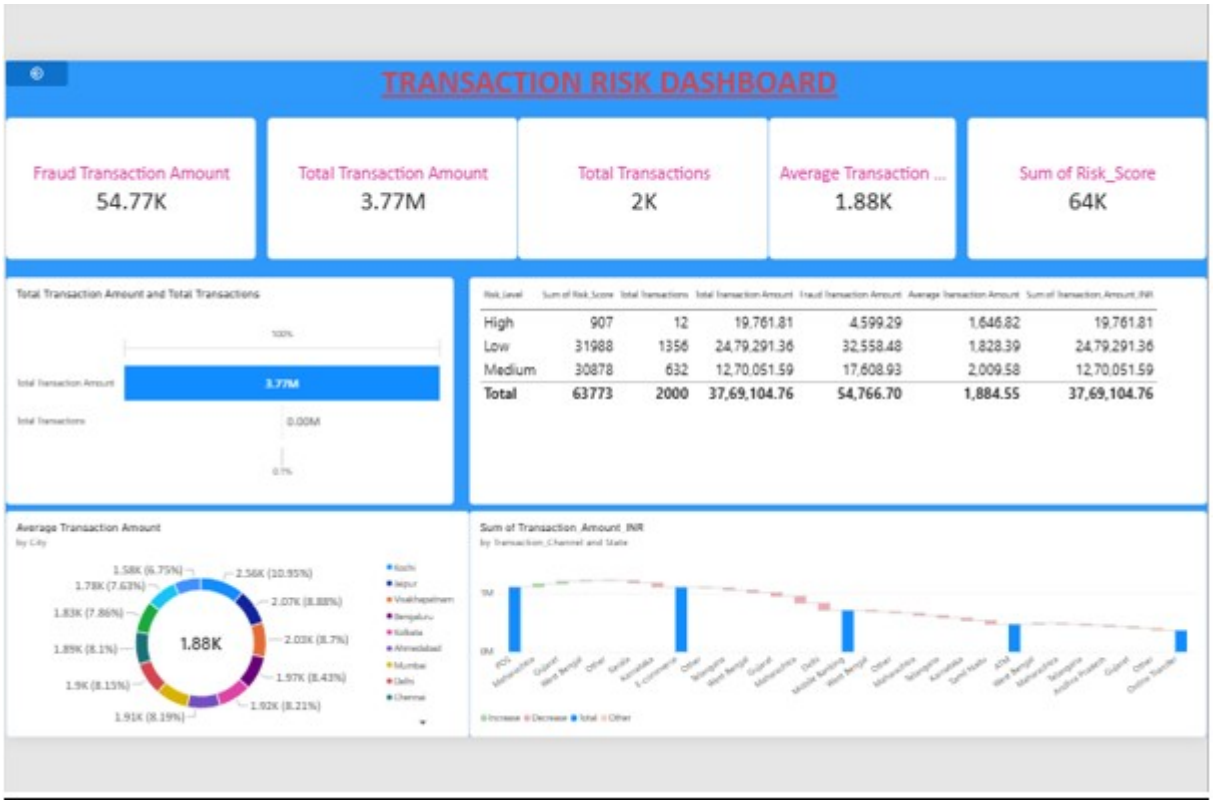
POWERBI DASHBOARD VISUALS:

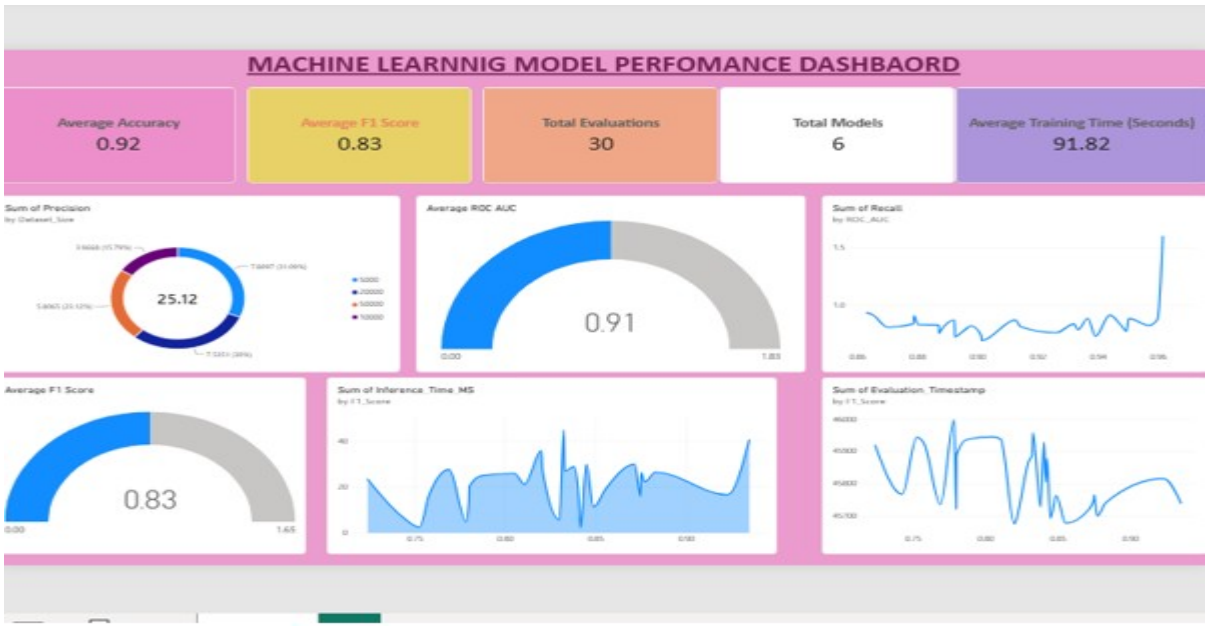


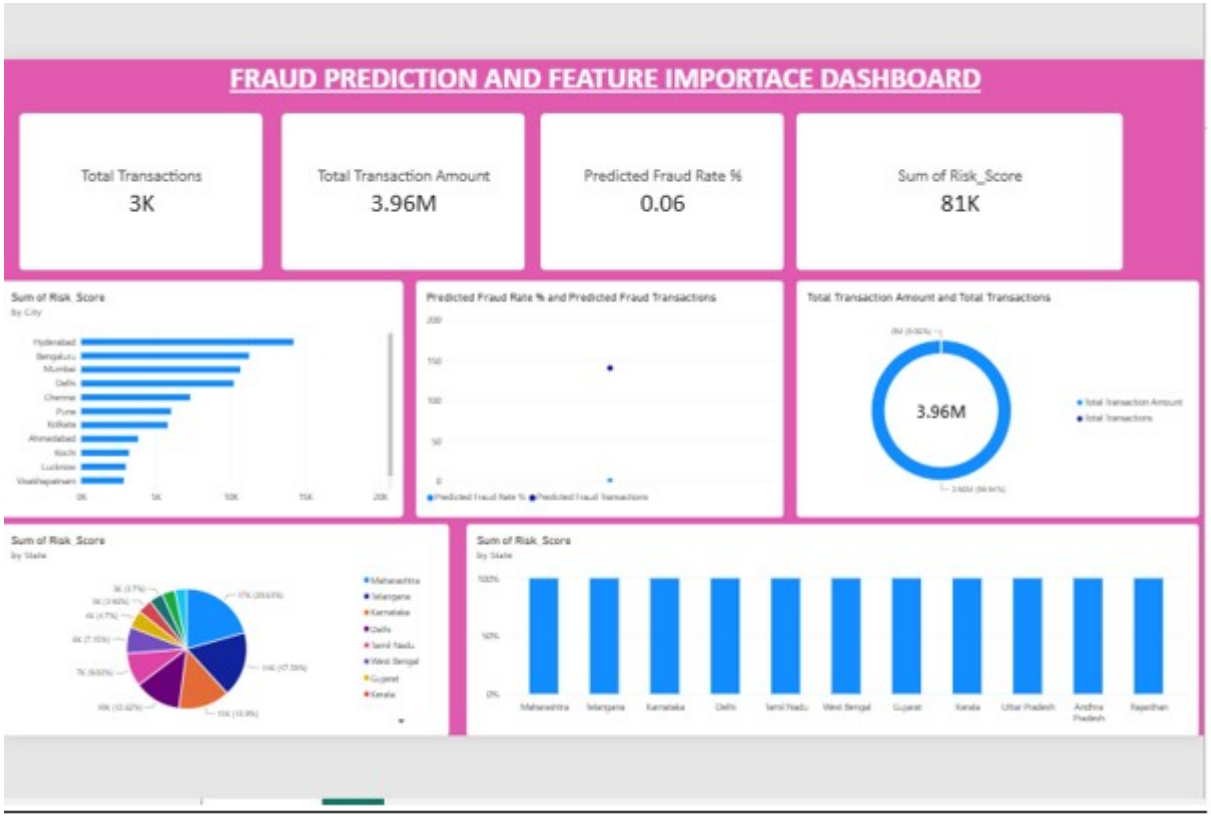


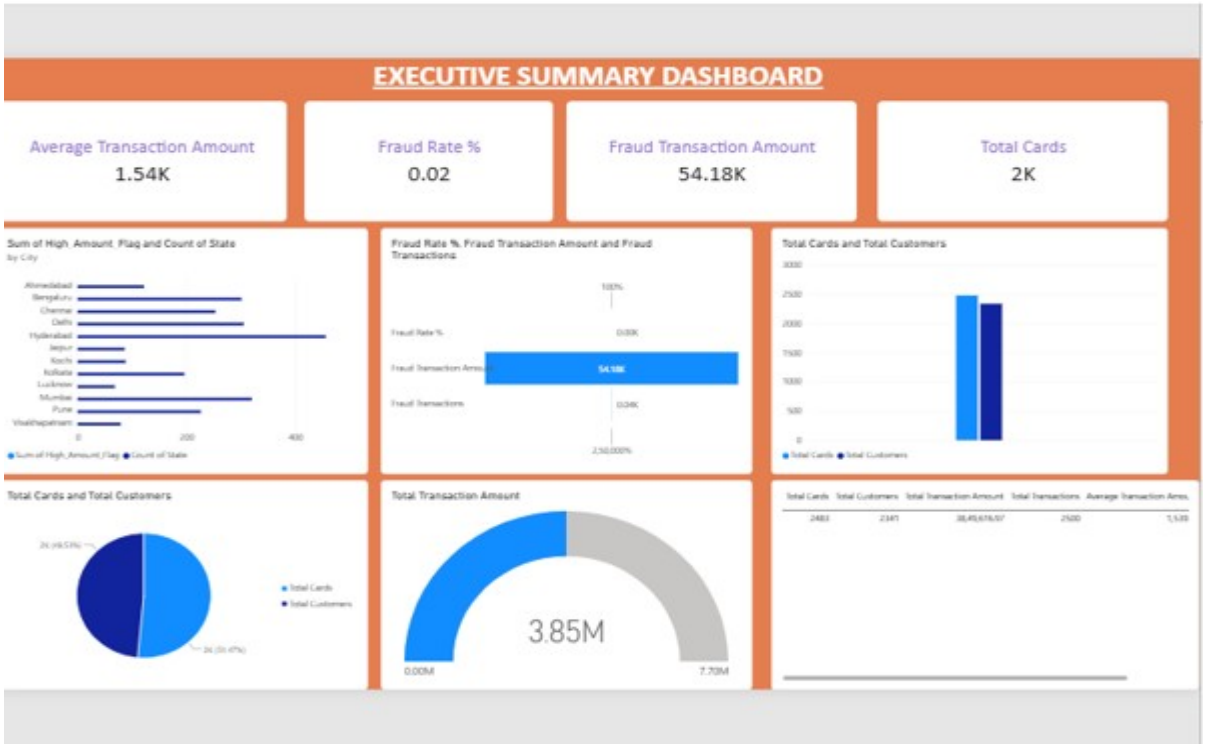

GEOGRAPHIC FRAUD DASHBOARD
Average Transaction Amount
1.60K
Legitimate Transactions
1K
Total Transaction Amount
2.40M
Total Transactions
2K
%GT Sum of Fraud_Rate_...
100.00%
Average Transaction Amount and Sum of Avg_Risk_Score
Fraud Amount, Fraud Rate (%) and Sum of Fraud_Amount_INR
177.46K
Legitimate Transactions and Total Transaction Amount
Sum of Avg_Risk_Score
Fraud Amount and Fraud Transactions
Legitimate Transactions

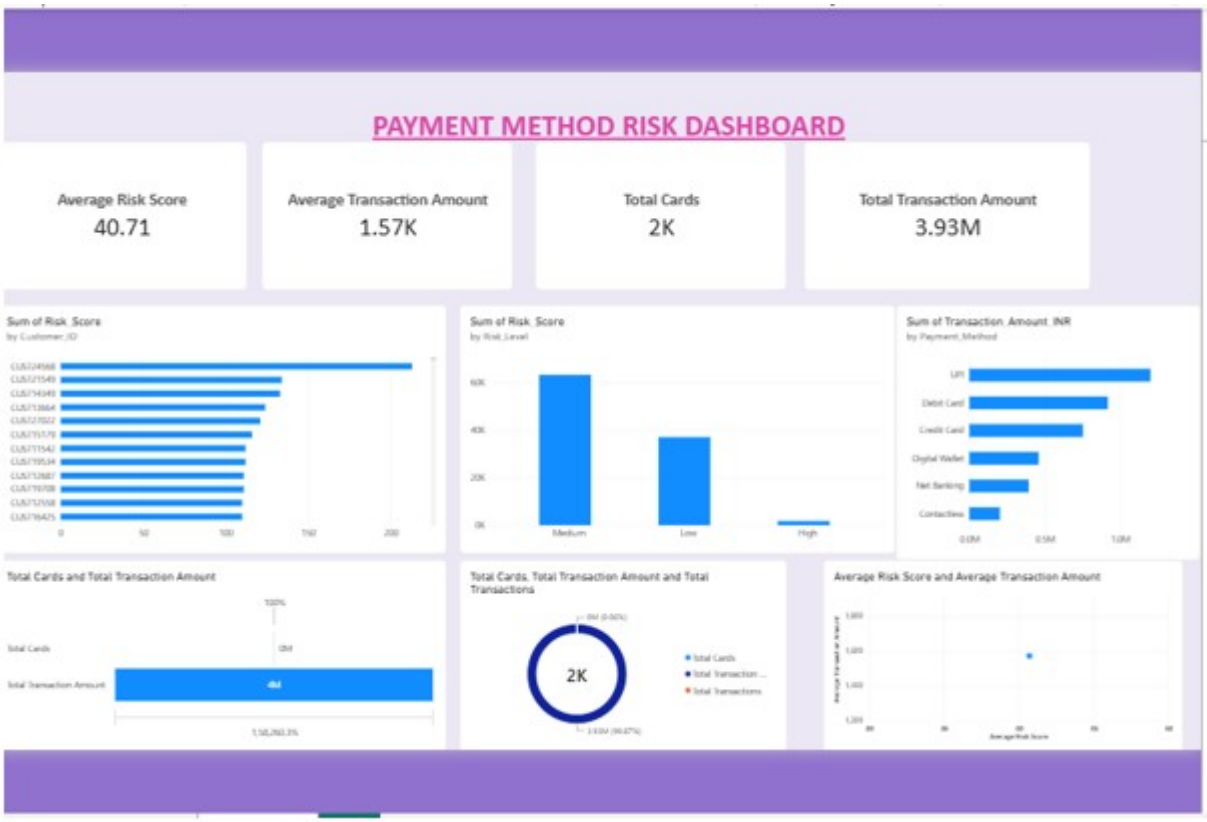

TRANSACRTION TREND DASHBOARD
Different Customers
2K
Fraud Transactions
75
Total Transaction Amount
3.04M
Average Transaction Amount
1.52K
Unique Cards
2K
Sum of Transaction_Timestamp
Unique Cards and Unique Customers
Fraud Transactions
Count of City
Count of Transaction_ID

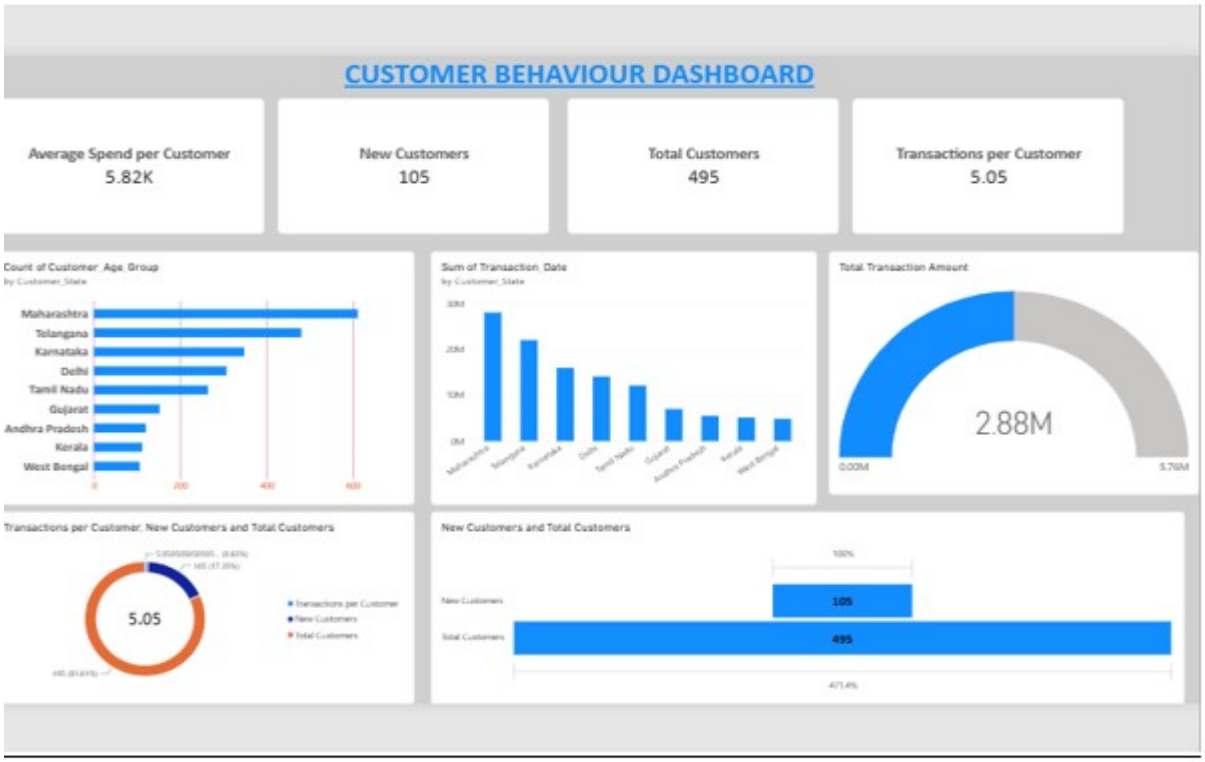

TRANSACTION RISK DASHBOARD
Fraud Transaction Amount
54.77K
Total Transaction Amount
3.77M
Total Transactions
2K
Average Transaction ...
1.88K
Sum of Risk_Score
64K
Total Transaction Amount and Total Transactions
3.77M
0.00M
Risk_Level | Sum of Risk_Score | Total Transactions | Total Transaction Amount | Fraud Transaction Amount | Average Transaction Amount | Sum of Transaction_Amount_INR
High | 907 | 12 | 19,761.81 | 4,599.29 | 1,646.82 | 19,761.81
Low | 31988 | 1356 | 24,79,291.36 | 32,558.48 | 1,828.39 | 24,79,291.36
Medium | 30878 | 632 | 12,70,051.59 | 17,608.93 | 2,009.58 | 12,70,051.59
Total | 63773 | 2000 | 37,69,104.76 | 54,766.70 | 1,884.55 | 37,69,104.76
Average Transaction Amount
by City
1.88K
Sum of Transaction_Amount_INR
by Transaction_Channel and State

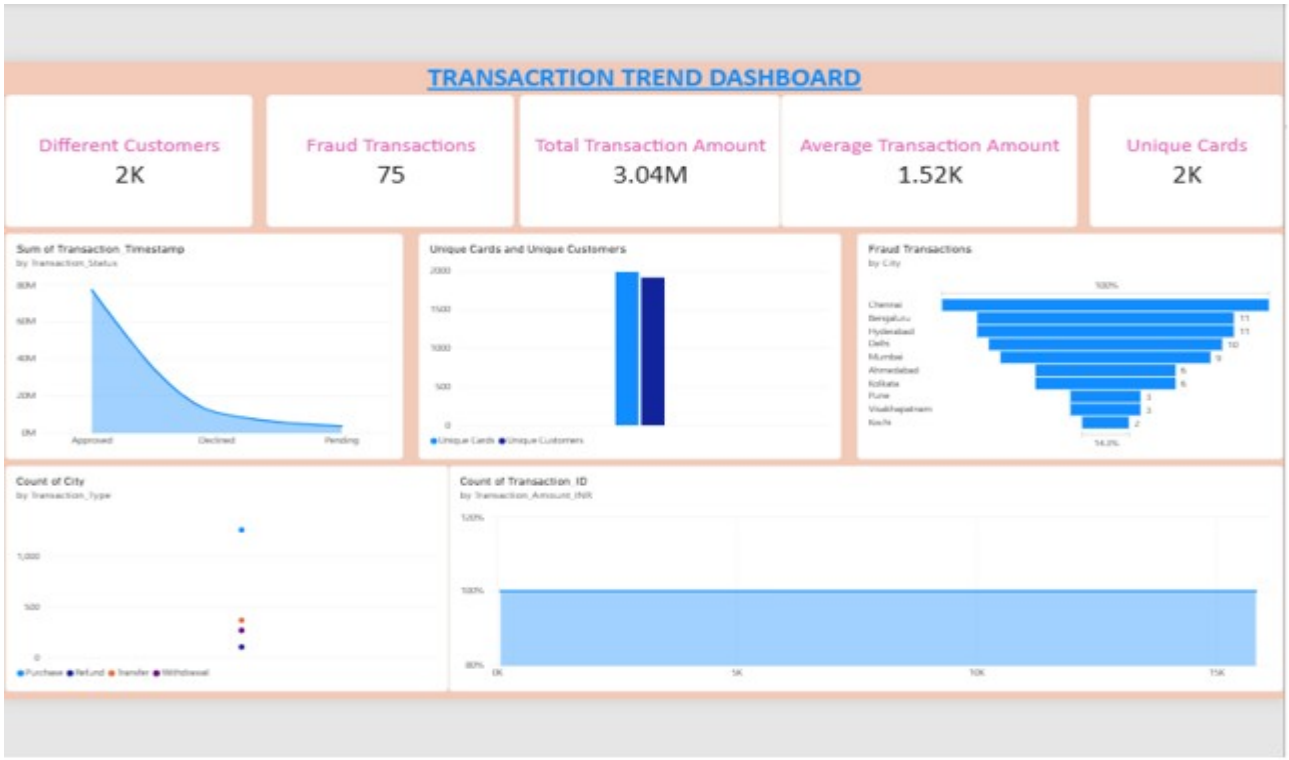

MACHINE LEARNNIG MODEL PERFOMANCE DASHBAORD
Average Accuracy
0.92
Average F1 Score
0.83
Total Evaluations
30
Total Models
6
Average Training Time (Seconds)
91.82
Sum of Precision
by Dataset_Size
25.12
Average ROC AUC
0.91
Sum of Recall
by ROC_AUC
Average F1 Score
0.83
Sum of Inference_Time_MS
by F1_Score
Sum of Evaluation_Timestamp
by F1_Score


FRAUD PREDICTION AND FEATURE IMPORTACE DASHBOARD
Total Transactions
3K
Total Transaction Amount
3.96M
Predicted Fraud Rate %
0.06
Sum of Risk_Score
81K
Sum of Risk_Score
by City
Predicted Fraud Rate % and Predicted Fraud Transactions
Total Transaction Amount and Total Transactions
3.96M
Sum of Risk_Score
by State
Sum of Risk_Score
by State

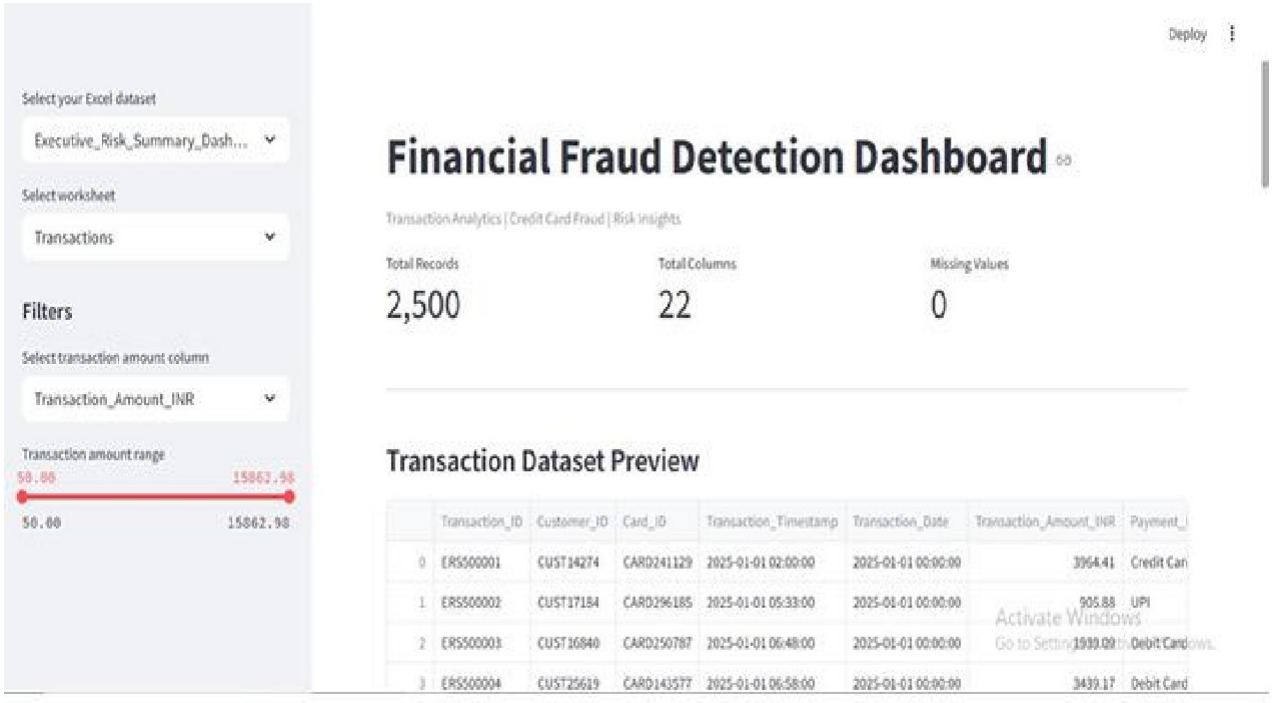

EXECUTIVE SUMMARY DASHBOARD
Average Transaction Amount
1.54K
Fraud Rate %
0.02
Fraud Transaction Amount
54.18K
Total Cards
2K
Sum of High_Amount_Flag and Count of State
by City
Fraud Rate %, Fraud Transaction Amount and Fraud Transactions
Total Cards and Total Customers
Total Cards and Total Customers
Total Transaction Amount
3.85M
0.00M
7.70M

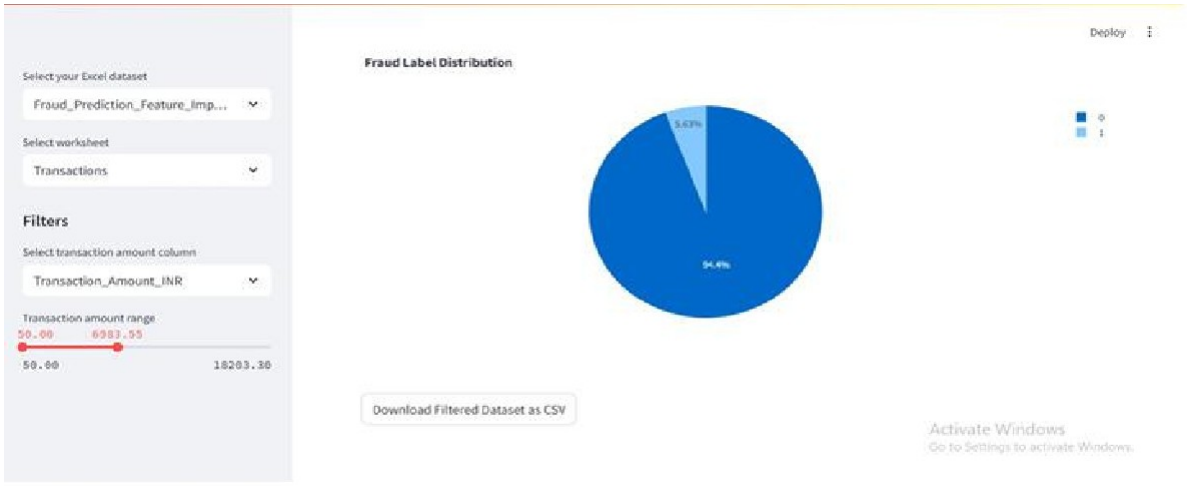

PAYMENT METHOD RISK DASHBOARD
Average Risk Score
40.71
Average Transaction Amount
1.57K
Total Cards
2K
Total Transaction Amount
3.93M
Sum of Risk_Score
Sum of Risk_Score
Sum of Transaction_Amount_INR
Total Cards and Total Transaction Amount
Total Cards, Total Transaction Amount and Total Transactions
2K
Average Risk Score and Average Transaction Amount

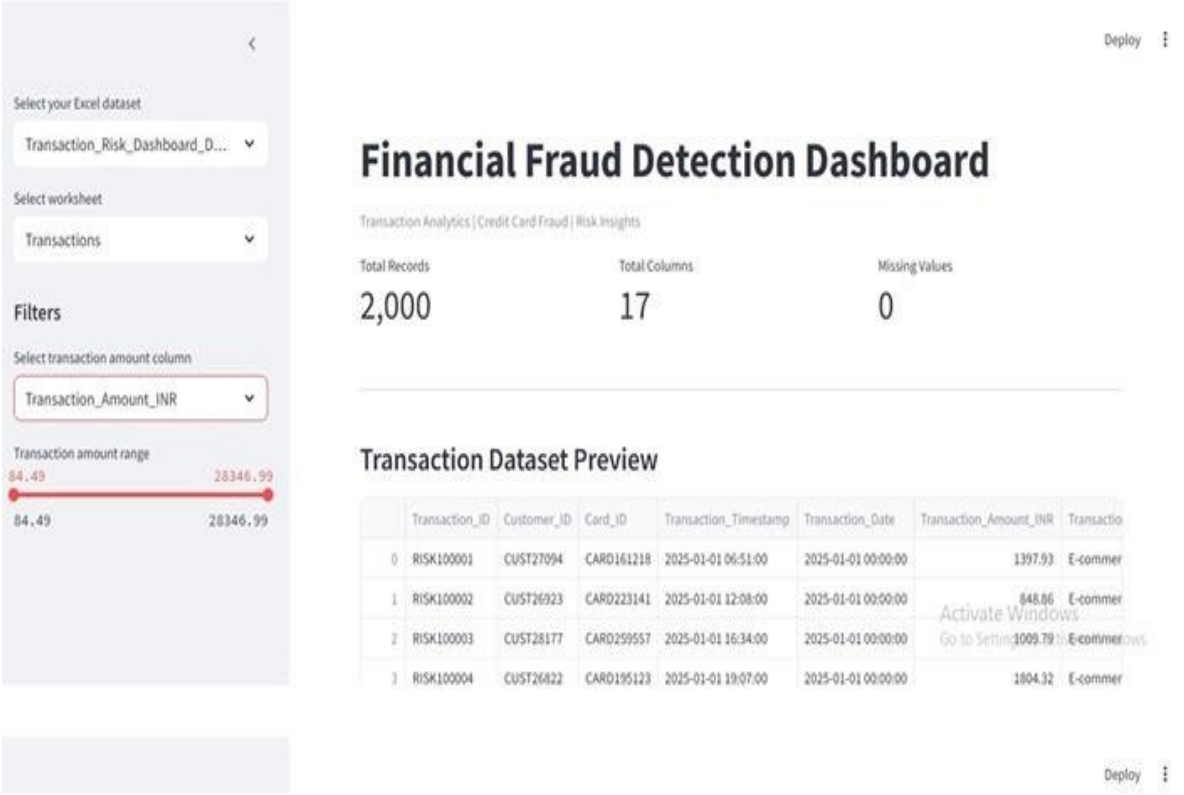

CUSTOMER BEHAVIOUR DASHBOARD
Average Spend per Customer
5.82K
New Customers
105
Total Customers
495
Transactions per Customer
5.05
Count of Customer_Age_Group
Maharashtra
Telangana
Karnataka
Delhi
Tamil Nadu
Gujarat
Andhra Pradesh
Kerala
West Bengal
Sum of Transaction_Date
Total Transaction Amount
2.88M
Transactions per Customer, New Customers and Total Customers
5.05
New Customers and Total Customers
105
495

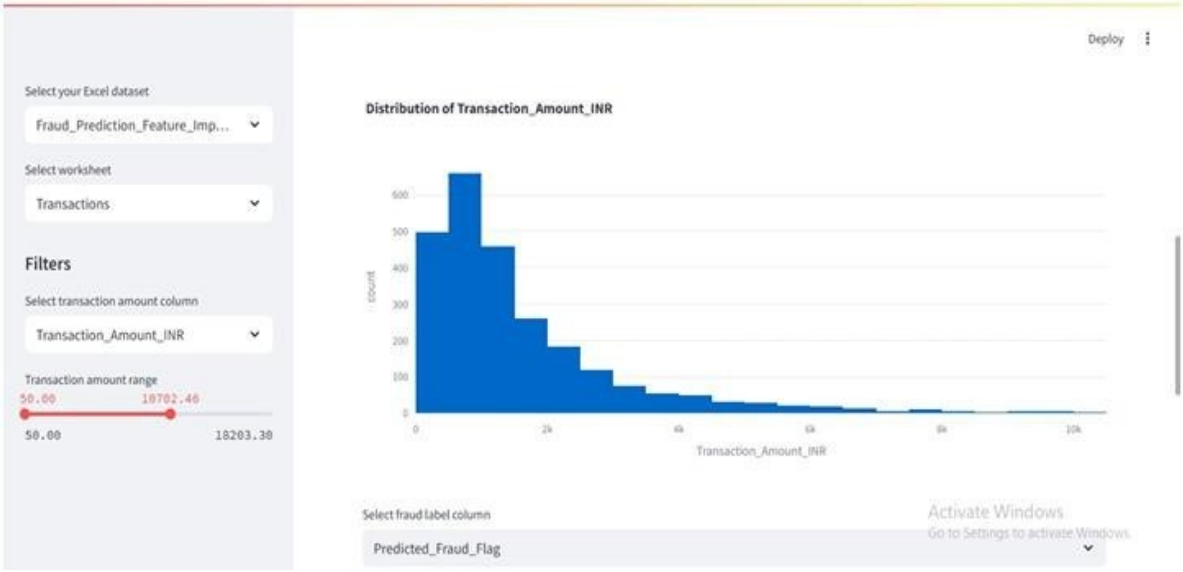


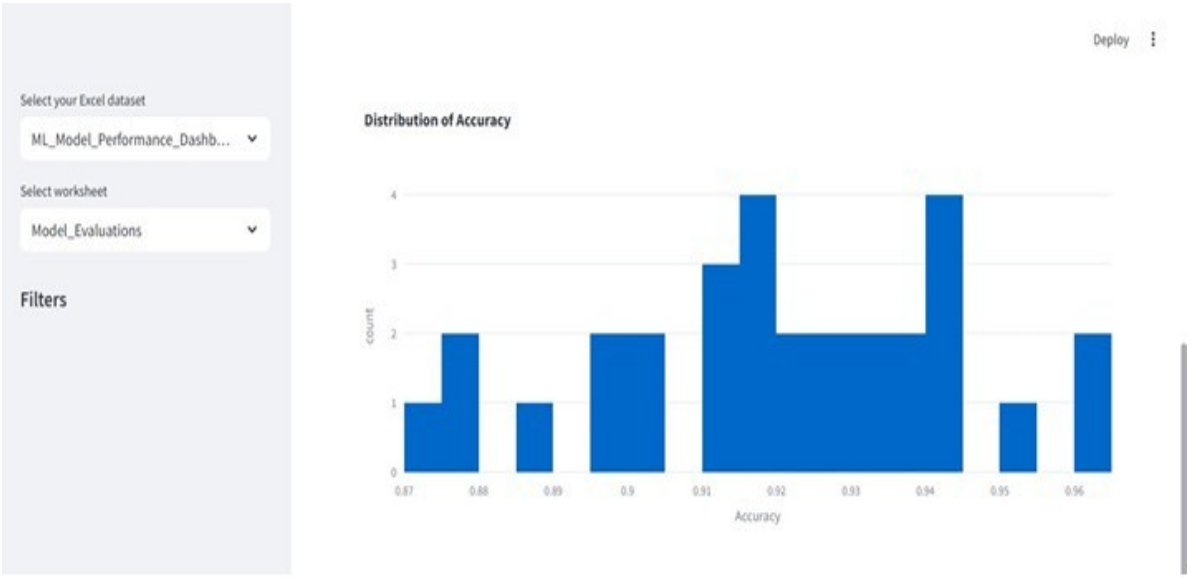
STREAMLIT DASHBOARD VISUALS:




Deploy
Select your Excel dataset
Fraud_Prediction_Feature_Imp...
Select worksheet
Transactions
Filters
Select transaction amount column
Transaction_Amount_INR
Transaction amount range
50.00
6983.55
50.00
18203.30
Fraud Label Distribution
5.63%
94.4%
Download Filtered Dataset as CSV
Activate Windows
Go to Settings to activate Windows.

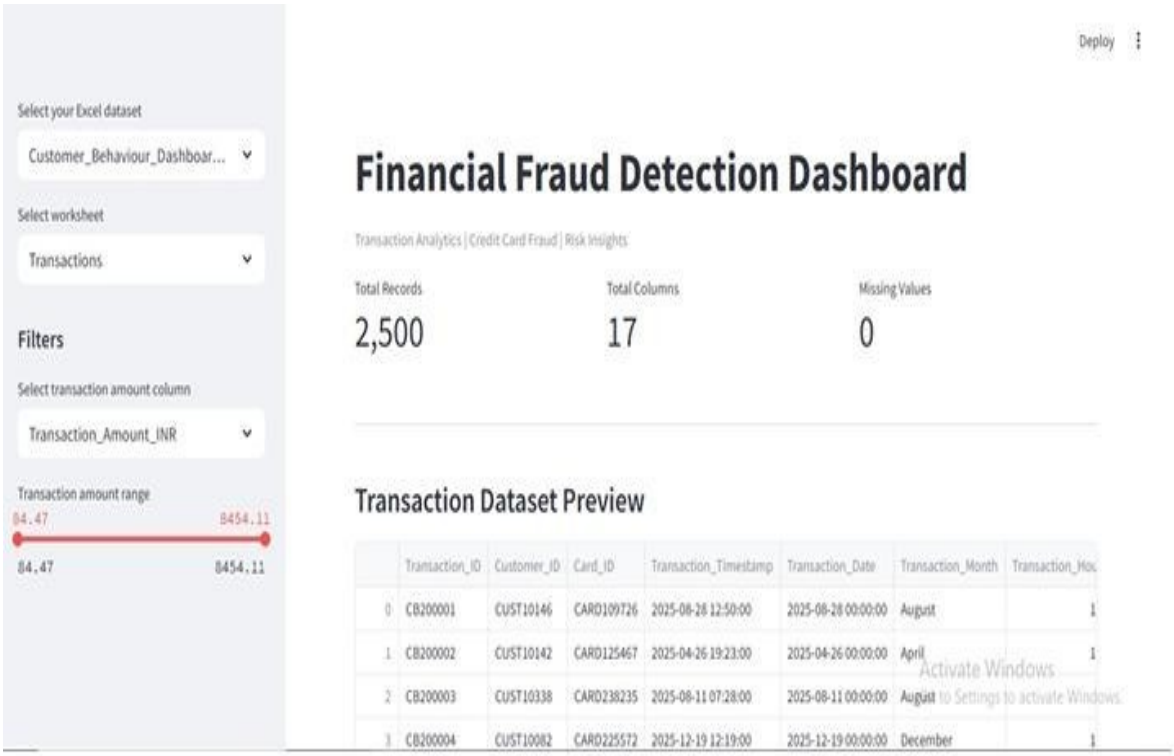

Deploy
Select your Excel dataset
Transaction_Risk_Dashboard_D...
Select worksheet
Transactions
Filters
Select transaction amount column
Transaction_Amount_INR
Transaction amount range
84.49
28346.99
84.49
28346.99
Financial Fraud Detection Dashboard
Transaction Analytics | Credit Card Fraud | Risk Insights
Total Records
2,000
Total Columns
17
Missing Values
0
Transaction Dataset Preview
Transaction_ID
Customer_ID
Card_ID
Transaction_Timestamp
Transaction_Date
Transaction_Amount_INR
Transactio
0 RISK100001 CUST27094 CARD161218 2025-01-01 06:51:00 2025-01-01 00:00:00 1397.93 E-commer
1 RISK100002 CUST26923 CARD223141 2025-01-01 12:08:00 2025-01-01 00:00:00 848.86 E-commer
2 RISK100003 CUST28177 CARD259557 2025-01-01 16:34:00 2025-01-01 00:00:00 1009.79 E-commer
3 RISK100004 CUST26822 CARD195123 2025-01-01 19:07:00 2025-01-01 00:00:00 1804.32 E-commer
Activate Windows

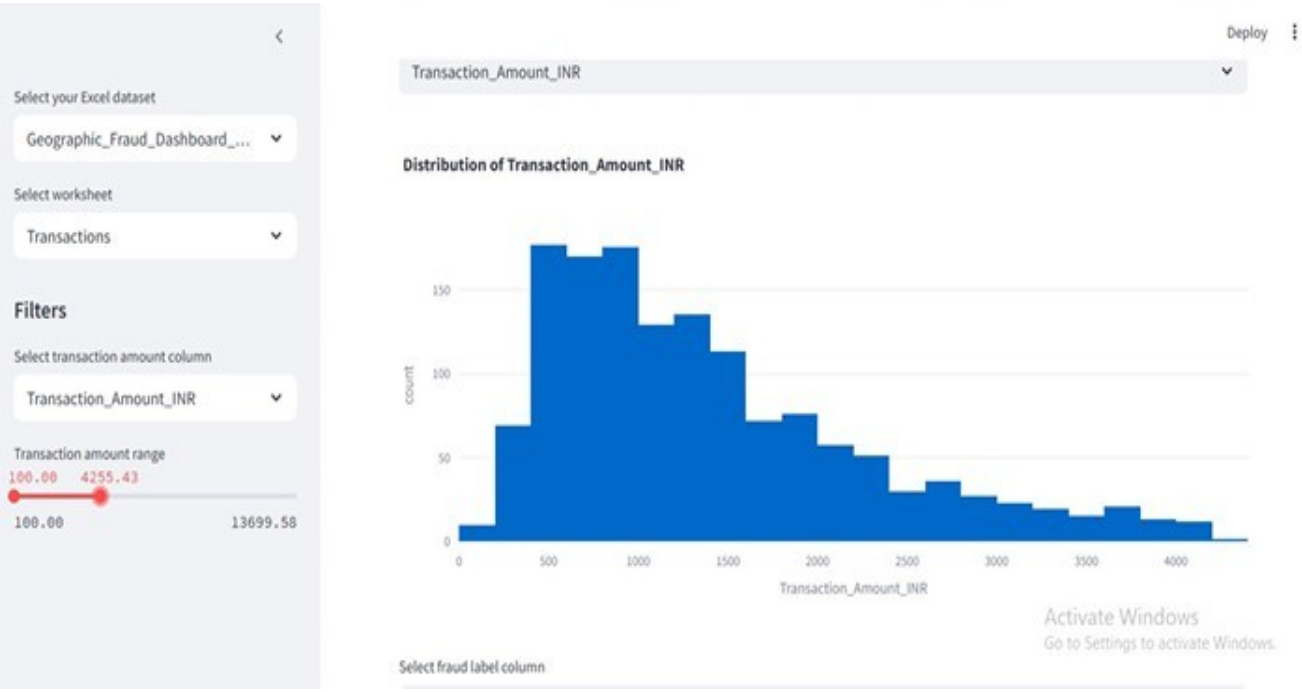

Deploy
Select your Excel dataset
Fraud_Prediction_Feature_Imp...
Select worksheet
Transactions
Filters
Select transaction amount column
Transaction_Amount_INR
Transaction amount range
50.00
18702.46
50.00
18203.30
Distribution of Transaction_Amount_INR
count
Transaction_Amount_INR
Select fraud label column
Predicted_Fraud_Flag
Activate Windows


Deploy
Select your Excel dataset
ML_Model_Performance_Dashb...
Select worksheet
Model_Evaluations
Filters
Distribution of Accuracy
count
Accuracy
0
1
2
3
4
0.87
0.88
0.89
0.9
0.91
0.92
0.93
0.94
0.95
0.96


Deploy
Transaction_Hour
Select your Excel dataset
Customer_Behaviour_Dashboar...
Select worksheet
Transactions
Filters
Select transaction amount column
Transaction_Amount_INR
Transaction amount range
84.47
8454.11
84.47
8454.11
Distribution of Transaction_Hour
count
Transaction_Hour
0
20
40
60
80
100
120
0
5
10
15
20
Activate Windows
Go to Settings to activate Windows.


Deploy
Select your Excel dataset
Fraud_Analysis_Dashboard_Dat...
Select worksheet
Transactions
Filters
Select transaction amount column
Amount_INR
Transaction amount range
199.00
99999.00
199.00
99999.00
Distribution of Amount_INR
count
Amount_INR
No standard fraud-label column was detected. We can map your dataset's actual fraud column in the next step.
Download Filtered Dataset as CSV
Activate Windows
Go to Settings to activate Windows.


Deploy
Select your Excel dataset
Customer_Behaviour_Dashboar...
Select worksheet
Transactions
Filters
Select transaction amount column
Transaction_Amount_INR
Transaction amount range
84.47
8454.11
84.47
8454.11
Financial Fraud Detection Dashboard
Transaction Analytics | Credit Card Fraud | Risk Insights
Total Records
2,500
Total Columns
17
Missing Values
0
Transaction Dataset Preview
Transaction_ID
Customer_ID
Card_ID
Transaction_Timestamp
Transaction_Date
Transaction_Month
0 CB200001 CUST10146 CARD109726 2025-08-28 12:50:00 2025-08-28 00:00:00 August 1
1 CB200002 CUST10142 CARD125467 2025-04-26 19:23:00 2025-04-26 00:00:00 April 1
2 CB200003 CUST10338 CARD238235 2025-08-11 07:28:00 2025-08-11 00:00:00 August 1
3 CB200004 CUST10082 CARD225572 2025-12-19 12:19:00 2025-12-19 00:00:00 December 1
Activate Windows
Go to Settings to activate Windows.



Project Output:

The Financial Fraud Detection Model with Dashboard analyzes credit-card transaction data using Python and machine learning techniques to identify patterns associated with fraudulent and legitimate transactions.

1. Data Analysis Output: Successfully loaded and examined the credit-card dataset containing 284,807 transactions and 31 columns**.**

2. Data Preprocessing Output: Prepared transaction data for analysis using preprocessing techniques, feature scaling, and class imbalance handling**.**

3. Machine Learning Output: Developed a workflow for analyzing and comparing Logistic Regression, KNN, SVC, Decision Tree, and Random Forest classification models.

4. Model Evaluation Output: Used evaluation metrics such as accuracy, precision, recall, F1-score, ROC-AUC, and confusion matrices to assess model

performance.

5. Fraud Detection Output: The classification workflow is designed to distinguish fraudulent transactions (Class = 1) from legitimate transactions (Class = 0).

6. Dashboard Output: The proposed Streamlit and Power BI dashboards can present fraud analysis, transaction summaries, and model evaluation results through interactive visualizations.

CONCLUSION OF THIS PROJECT:

- The Financial Fraud Detection Model with Dashboard demonstrates the application of Machine Learning and Data Analytics in identifying potentially fraudulent credit-card transactions.

- The project uses Python libraries such as Pandas, NumPy, Matplotlib, Seaborn, and Scikit-learn to perform data preprocessing, exploratory data analysis, model development, and performance evaluation.

- The project also outlines the use of Streamlit and Power BI dashboards to present transaction patterns, fraud analysis, and model performance in an interactive and understandable format.

- Overall, this project provides a foundation for machine learning-based financial fraud analysis and data-driven decision-making.

- Future enhancements include real-time transaction monitoring, automated fraud alerts, advanced anomaly detection, and continuous model improvement to strengthen fraud detection capabilities.